Online Appendix to
Publication Bias and Model Uncertainty in Measuring the Effect of Class Size on Achievement*

Matej Opatrny[a], Tomas Havranek[a,b,c], Zuzana Irsova[a] , Milan Scasny[a]

[a]Charles University, Prague
[b]Centre for Economic Policy Research, London
[c]Meta-Research Innovation Center, Stanford

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Abstract

Class size reduction mandates are routinely justified by studies reporting positive effects on student achievement. Yet other studies report no effects, and the literature as a whole awaits correction for potential publication bias. Moreover, if identification drives results systematically, the relevance of individual studies will vary. We build a sample of 2,819 estimates collected from 66 studies and for each estimate classify 42 factors that reflect estimation context. We employ nonlinear techniques for publication bias correction and model averaging techniques to address model uncertainty. The results are consistent with little publication bias. The implied class size effect is negligible for all identification approaches except Tennessee's Student/Teacher Achievement Ratio project and for all contexts except classes of fewer than 15 students.

*Data, code, and the main body of the paper are available at `meta-analysis.cz/class`. Corresponding author: Zuzana Irsova, `zuzana.irsova@ies-prague.org`.

Appendices

B Online Appendix

Table B1: Jurisdictions with class size reductions since 2010

Jurisdiction	Year	Description	Source
Australia (New South Wales)	2016	Agreement between the government and teachers union to have on average 20 students in grade K, 22 in grade 1, 24 in grade 2, 30 in grades 3–10, 24 in grades 11–12.	NSWTF (2019) NSWG (2023)
Canada (Ontario)	2012	Ontario regulation 132/12, average of 26 students to 2 instructors in kindergarten, cap of 23 in grades K–3, average of 24.5 students in grades 4–8, and average of 22 in grades 9–12.	OME (2019)
Canada (Quebec)	2011	Focus on underprivileged areas with cap at 20 students for grades 3–4, and at 24 students for grades 5–6.	QME (2023)
Finland	2010 2012	In the quality criteria of 2010 and 2012 by the Finnish Ministry for Education and Culture recommendation on class size of 20–25 pupils for grades 1–6.	FNAE (2019) MECF (2012)
France	2017	In 2017 addendum to the Education Law, underprivileged areas cap set at 12 students in grade 1, in 2018 extended to grade 2 (it followed previous experiment in 2002-2003 where in underprivileged areas, cap was 10 students in grade 1). In 2020, in all areas the class size cap at 24 students for grades K–3.	Bressoux *et al.* (2019) Evain (2022)
Germany (Hesse)	2011	In 2011 Hessian Education state law amendment, class sizes reduced for primary schools to a maximum of 25 students.	Argaw & Puhani (2018)
India	2021	In 2021 the Indian New Education policy reducing the student-to-teacher ratio to 25:1 for primary schools and 30:1 for upper primary schools by 2022.	NIE (2020)
Israel	2015	In 2015, the Israeli government approved of a plan to cap grades 1–2 to no more than 34 students per class by 2020.	JP (2015)
New Zealand	2023	In 2023, the NZ Minister of Education announced that student-to-teacher ratios for grades 4–8 will be reduced from 29:1 to 28:1 by 2025.	Tinetti (2023)
Norway	2017 2019	In 2017 Norwegian parliament voted on upper limit on student-to-teacher ratio 16:1 in grades 1–4 and 21:1 in grades 5–10. In 2019 these ratios were reduced to 15:1 in grades 1–4 and 20:1 in grades 5–10.	MERN (2019)
Portugal	2017	In 2017, the Portuguese government announced a class size reduction policy of average class size of 20 students in primary schools and 26 students in secondary schools by 2021.	OECD (2020)
South Korea	2015	The Ministry of Education announced plan to reduce average class size from 30 to 24 and student/teacher ratio from 16.6 to 13.3 till 2022.	Han & Ryu (2017) Koreaherald (2016)
Spain (Madrid)	2020	Cap of 20 students for grades 1–3 due to covid-19 pandemics regionally.	Elpais (2020)
United Kingdom (Scotland)	2010	Cap reduced to 25 students for grade 1 and composite age classes, and from 30 to 33 students for other grades.	NASUWT (2023)
United States (California)	2013 2022	In 2013, class size cap of 24 students in grades K–3, in 2022 Senate Bill 1431 lowered student-to-teacher ratio to 20:1 in grades K–3.	Rubio (2022)
United States (New York City)	2022	In 2022 NY state senate bill S9460, class size cap of 20 students in grades K–3, of 23 students in grades 4–8, and of 25 students per class in high school.	Fadulu (2022)
United States (Wisconsin)	2015	In the follow-up of the SAGE program in 2015 (Wisconsin Acts 53 and 71, Achievement Gap Reduction program), participating schools have to reduce student-to-teacher ratio to max 18:1 or 30:2.	WDPI (2016) VARC (2023)

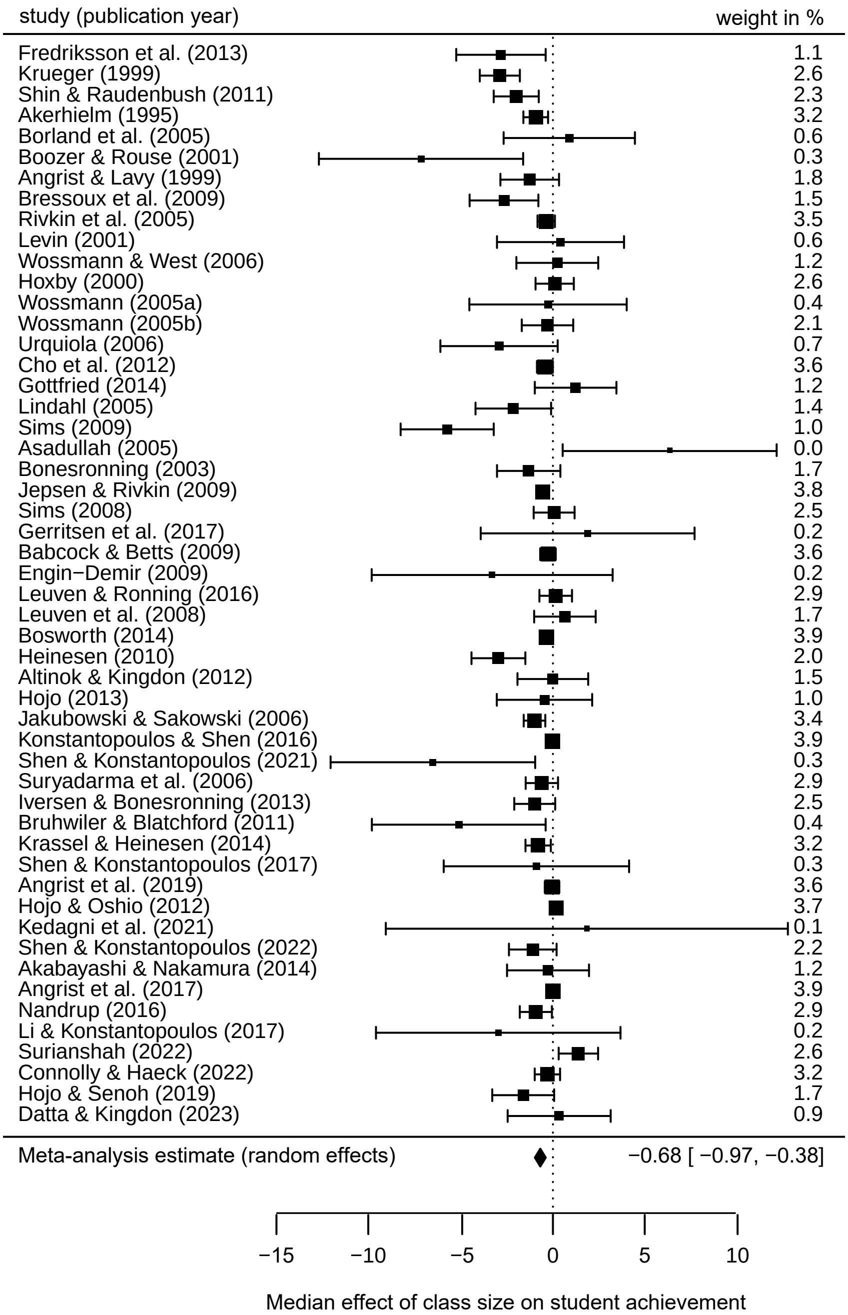
Notes: The table gives examples on class size reduction mandates or recommendations (accompanied by additional government funding) in various regions since 2010. The list is not exhaustive. Prior to 2010, at least 24 US states started to mandate or incentivize reductions (Whitehurst & Chingos, 2011).

Table B2: Summary statistics for subsets of literature based on preferred estimates

		Unweighted			Weighted		
	Observations	Mean	95% conf. int.		Mean	95% conf. int.	
Subjects tested							
Math	290	-0.59	-0.89	-0.29	-1.42	-1.74	-1.10
Reading	64	-1.32	-1.91	-0.73	-1.91	-2.61	-1.21
Writing	9	-0.79	-1.69	0.11	-0.93	-1.61	-0.25
Languages	41	-0.43	-0.93	0.06	-1.63	-2.21	-1.05
Other subjects	133	-0.04	-0.39	0.31	-1.18	-1.89	-0.47
Class and student characteristics							
Kindergarten	25	-0.76	-1.23	-0.28	-0.91	-1.49	-0.33
Primary school	284	-0.86	-1.15	-0.56	-0.81	-1.08	-0.54
Secondary school	144	-0.31	-0.74	0.12	-2.09	-2.72	-1.47
Disadvantaged students	35	-1.18	-2.41	0.05	-1.10	-2.28	0.07
General population	348	-0.47	-0.70	-0.24	-1.27	-1.59	-0.94
Data characteristics							
Longitudinal data	30	-0.81	-1.46	-0.17	-2.07	-3.12	-1.02
Cross-sectional data	412	-0.64	-0.89	-0.39	-1.06	-1.36	-0.76
United States	89	-0.77	-1.10	-0.45	-0.81	-0.97	-0.66
Scandinavian countries	45	-0.92	-1.77	-0.07	-1.25	-1.83	-0.67
Other countries	310	-0.58	-0.88	-0.28	-1.50	-1.91	-1.10
Estimation characteristics							
STAR experiment	20	-2.05	-2.43	-1.66	-2.48	-2.84	-2.12
Regression discontinuity	75	-0.42	-0.63	-0.21	-0.53	-0.76	-0.29
Instrumental variable	228	-0.96	-1.37	-0.54	-1.31	-1.82	-0.79
Fixed effects	115	0.17	-0.08	0.42	0.16	-0.12	0.44
Quasi-experimental	438	-0.62	-0.85	-0.38	-0.91	-1.20	-0.62
OLS	6	-3.08	-5.56	-0.61	-3.47	-5.31	-1.63
Publication characteristics							
Top 5 journals in economics	32	-0.91	-1.38	-0.44	-2.01	-2.49	-1.54
Other than Top 5	412	-0.63	-0.88	-0.38	-1.16	-1.46	-0.86
All preferred estimates	444	-0.65	-0.89	-0.42	-1.22	-1.50	-0.93

Notes: The table shows subsample-specific means for the estimated effects of class size on achievement that are emphasized by the authors of the original studies. The effects are normalized to represent the change in hundredths of a standard deviation in test scores corresponding to an increase in class size by one student. That is, an estimate of -1 means that a class size reduction by 10 students is associated with an improvement in test scores by 0.1 standard deviations. In the left-hand portion of the table each estimate has the same weight. In the right-hand portion of the table each study has the same weight; in other words, there we weight estimates by the inverse of the number of estimates reported per study. Estimates are winsorized at the 1% level. For the definition of subsamples see Table 6. The group labeled "quasi-experimental" in the table also includes results for the STAR experiment and fixed effects.

Figure B1: Forest plot (preferred estimates)



Notes: From the sample of results preferred by the authors of each study the figure takes the median estimate and standard error. Square size denotes the weight of the study in meta-analysis; horizontal lines with bars represent 95% confidence intervals. The overall random effects meta-analysis mean and its confidence interval are displayed at the bottom.

Table B3: Class size effect for different size quantiles

Panel A: Unweighted

Quantile	N	Mean eff.	SD	Mean size	Min size	Max size
1	504	-0.99	1.96	18.7	12.7	20.4
2	616	-0.26	1.69	22.1	20.5	23.0
3	341	-0.08	2.24	24.3	23.2	25.6
4	532	-0.15	2.16	28.4	25.6	32.3
5	441	0.33	2.13	35.7	32.5	55.9
Total	2,434	-0.25	2.06	25.8	12.7	55.9

Panel B: Weighted by inverse variance

Quantile	N	Mean eff.	SD	Mean size	Min size	Max size
1	504	-0.02	0.14	18.7	12.7	20.4
2	616	-0.17	0.43	22.1	20.5	23.0
3	341	-0.05	0.28	24.3	23.2	25.6
4	532	-0.22	0.52	28.4	25.6	32.3
5	441	0.06	0.95	35.7	32.5	55.9
Total	2,434	-0.05	0.25	25.8	12.7	55.9

Notes: The table shows mean class size effects reported for different class sizes. The effects are normalized to represent the change in hundredths of a standard deviation in test scores corresponding to an increase in class size by one student. Specifically, an estimate of -1 implies that a class size reduction by 10 students is associated with an improvement in test scores by 0.1 standard deviations. In Panel B the reported effects of class size are weighted by inverse variance. SD = standard deviation, N = number of observations.

Figure B2: Distribution of class size

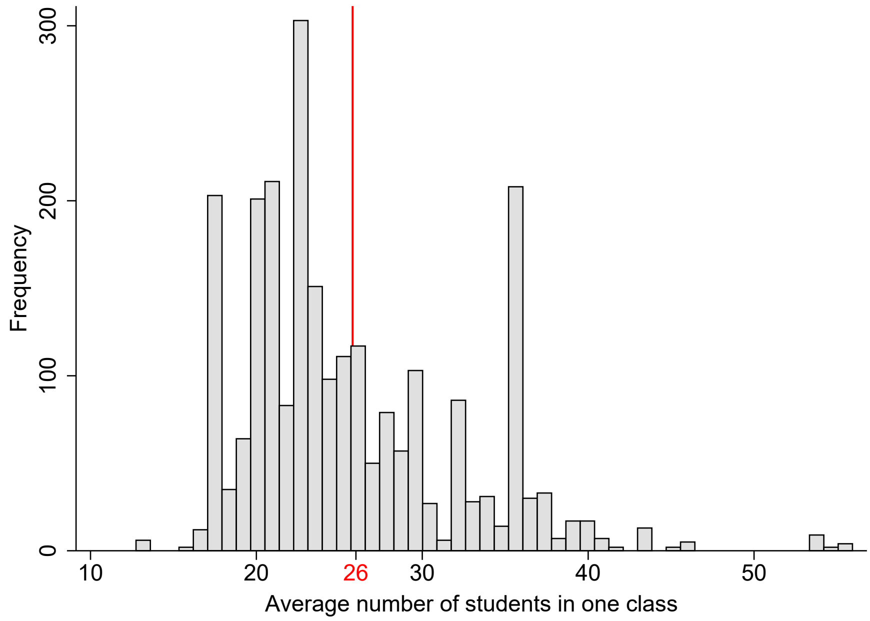


Figure B3: Nonlinearity in class size effects

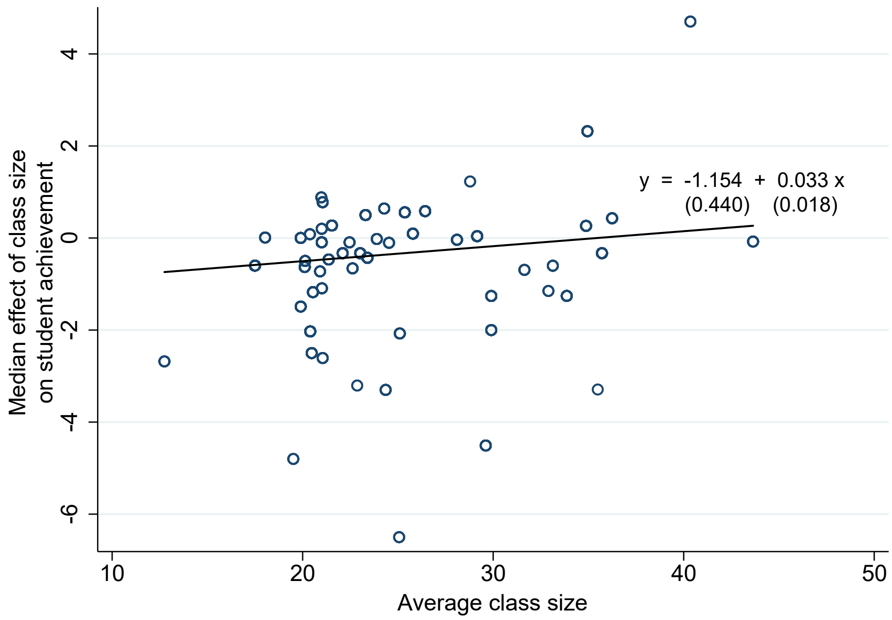
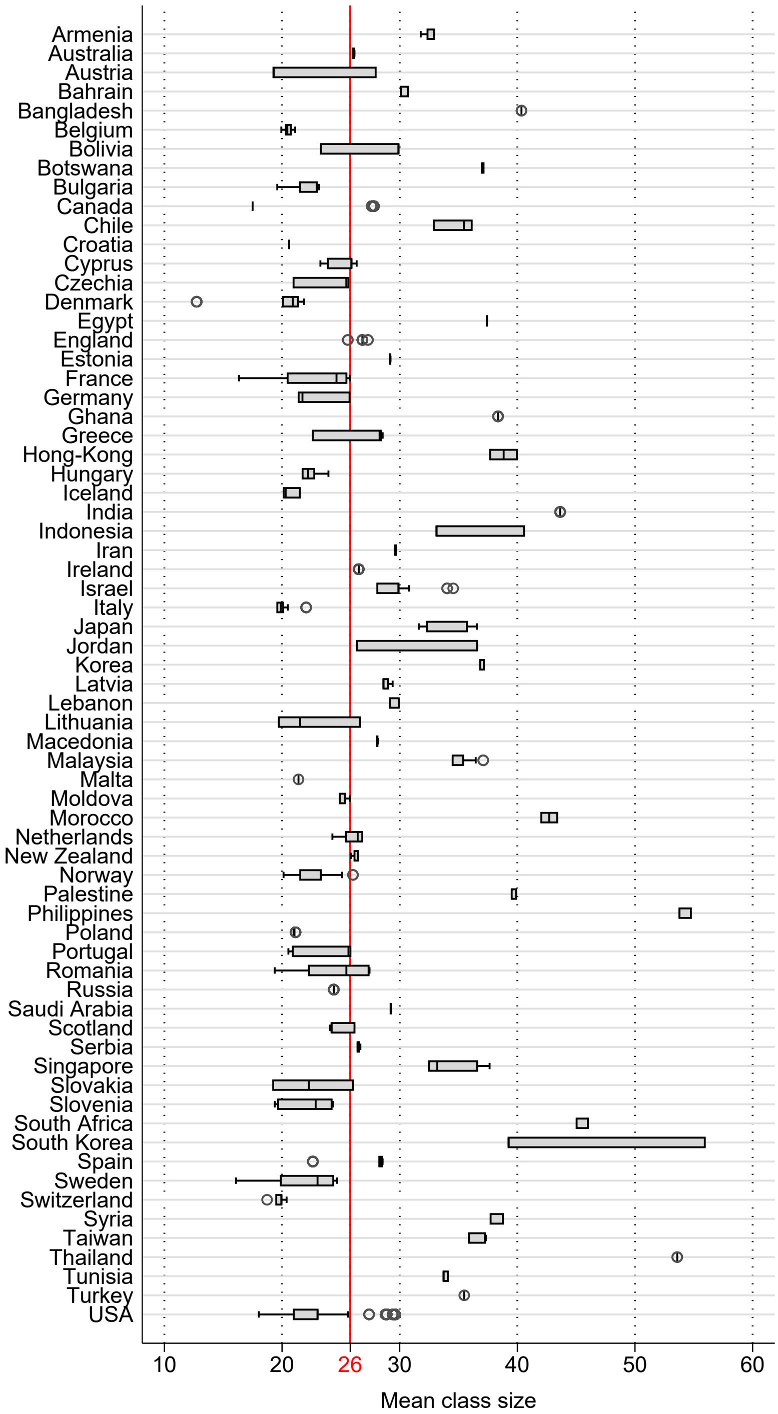
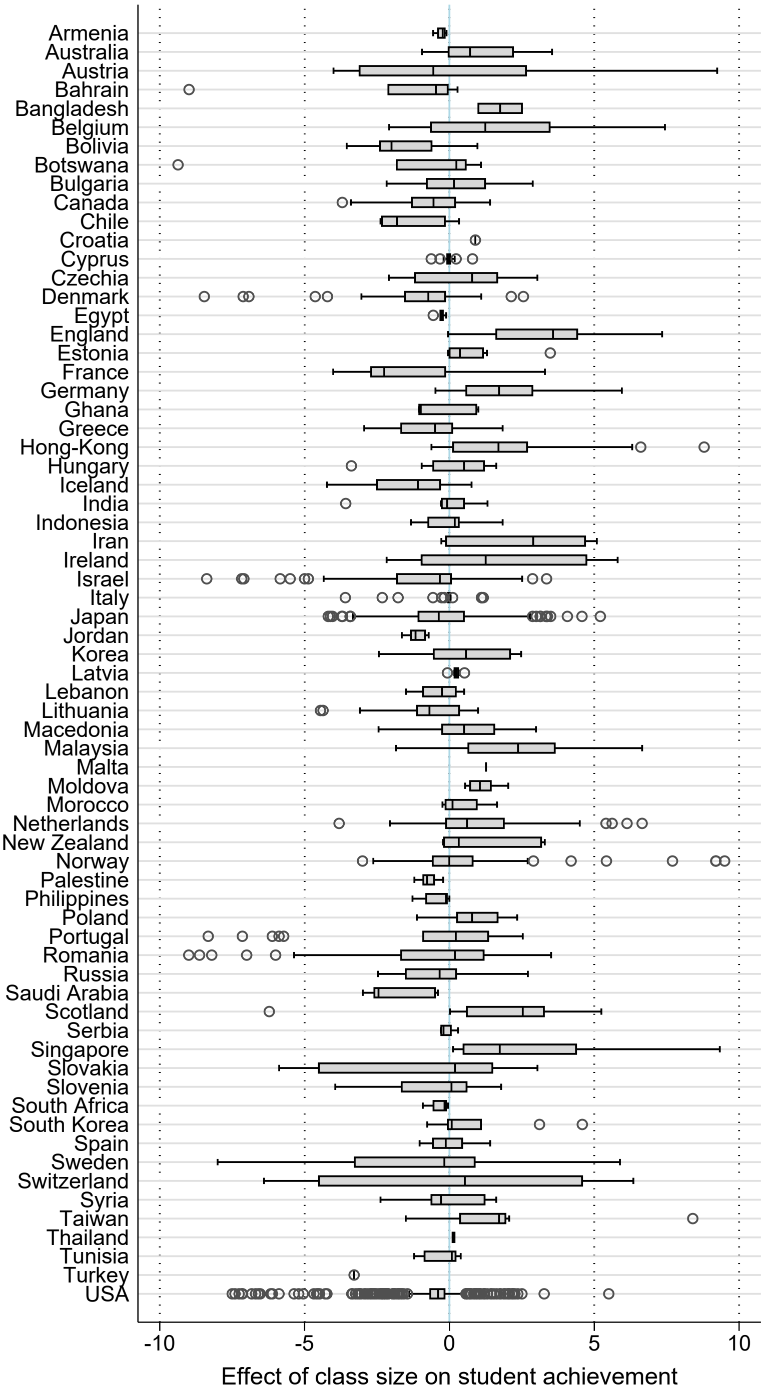


Figure B4: Class size differences within and across countries



Notes: The length of each box represents the interquartile range (P25-P75), and the line inside the box represents the median. The whiskers represent the smallest and largest estimates within 1.5 times the range between the upper and lower quartiles. Circles denote outliers. The solid vertical line denotes the overall mean (26).

Figure B5: Estimated class size effects vary within and across countries



Notes: The figure shows a box plot of the estimated effects of class size on achievement. The effects are normalized to represent the change in hundredths of a standard deviation in test scores corresponding to an increase in class size by one student. That is, an estimate of -1 means that a class size reduction by 10 students is associated with an improvement in test scores by 0.1 standard deviations. The length of each box represents the interquartile range (P25-P75), and the line inside the box represents the median. The whiskers represent the smallest and largest estimates within 1.5 times the range between the upper and lower quartiles. Circles denote outliers. Extreme outliers are excluded from the figure for ease of exposition but included in all statistical tests.

Table B4: Publication bias tests for subsets of data

Block 1: STAR experiment					
Panel A: Linear	OLS	FE	IV	Study	Precision
Publication bias	0.658	-0.0691	2.170**	1.436**	0.555
(*standard error*)	(0.732)	(0.743)	(0.923)	(0.721)	(0.638)
	[-0.622, 2.104]		[0.336, 4.004]	[-0.058, 3.016]	[-0.732, 2.083]
			{0.693, 3.994}		
Effect beyond bias	-2.407***	-1.943***	-3.374***	-3.138***	-2.342***
(*constant*)	(0.479)	(0.485)	(0.600)	(0.436)	(0.381)
	[-3.310, -1.679]		[-4.500, -2.250]	[-4.193, -2.225]	[-3.274, -1.590]
First-stage robust F-stat			112.8		
Panel B: Nonlinear	WAAP	Stem	Kink	p-uniform*	Selection
Publication bias			0.332		P = 1.986
			(0.660)		(1.769)
Effect beyond bias	-2.058***	-1.840***	-2.207***	-2.092***	-2.114***
	(0.106)	(0.287)	(0.388)	(0.112)	(0.195)
Observations	56	56	56	56	56

Block 2: Regression discontinuity					
Panel A: Linear	OLS	FE	IV	Study	Precision
Publication bias	0.0971	0.285	0.128	-0.195	-0.526
(*standard error*)	(0.334)	(0.294)	(0.556)	(0.213)	(0.443)
	[-1.598, 1.182]		[-2.714, 7.856]	[-1.772, 1.504]	[-1.829, 0.526]
			$\{-\infty, \infty\}$		
Effect beyond bias	-0.716***	-0.873***	-0.741***	-0.840***	-0.196
(*constant*)	(0.134)	(0.245)	(0.264)	(0.224)	(0.169)
	[-0.947, -0.341]		[-253.3, 101]	[-1.302, -0.344]	[-0.979, 1.079]
First-stage robust F-stat			37.5		
Panel B: Nonlinear	WAAP	Stem	Kink	p-uniform*	Selection
Publication bias			-1.358***		P = 1.171
			(0.145)		(0.438)
Effect beyond bias	-0.025***	-0.045	0.005	-0.633**	-0.548***
	(0.008)	(0.05)	(0.015)	(0.306)	(0.139)
Observations	436	436	436	436	436

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Table B4: Publication bias tests for subsets of data—cont.

Block 3: Instrumental variable					
Panel A: Linear	OLS	FE	IV	Study	Precision
Publication bias	-0.0689	-0.0689	-0.260	0.0309	-0.184
(*standard error*)	(0.194)	(0.154)	(0.311)	(0.291)	(0.194)
	[-0.756, 0.354]		[-0.601, 0.081]	[-0.865, 0.782]	[-0.640, 0.270]
			{-1.583, 0.386}		
Effect beyond bias	-0.272	-0.272	0.0497	-0.874*	-0.0783
(*constant*)	(0.227)	(0.259)	(0.431)	(0.458)	(0.115)
	[-0.720, 0.294]		[-0.497, 0.596]	[-1.849, 0.086]	[-0.373, 0.195]
First-stage robust F-stat			6.3		
Panel B: Nonlinear	WAAP	Stem	Kink	p-uniform*	Selection
Publication bias			-0.324***		P = 1.047
			(0.069)		(0.239)
Effect beyond bias	-0.005	0.03	0	-0.516	-0.238***
	(0.007)	(0.106)	(0.015)	(0.365)	(0.053)
Observations	845	845	845	845	845

Block 4: Fixed effects					
Panel A: Linear	OLS	FE	IV	Study	Precision
Publication bias	0.0955***	0.0685***	0.203***	0.217**	0.124**
(*standard error*)	(0.0106)	(0.00404)	(0.0606)	(0.104)	(0.0590)
	[-0.312, 0.405]		[-0.352, 0.758]	[0.036, 0.676]	[-0.748, 0.469]
			{0.077, 0.377}		
Effect beyond bias	-0.180	-0.146***	-0.337***	-0.322	-0.215***
(*constant*)	(0.114)	(0.00502)	(0.0445)	(0.298)	(0.0415)
	[-0.654, 0.085]		[-0.588, -0.087]	[-1.109, 0.302]	[-0.492, 0.028]
First-stage robust F-stat			9.2		
Panel B: Nonlinear	WAAP	Stem	Kink	p-uniform*	Selection
Publication bias			-0.391*		P = 0.774
			(0.207)		(0.177)
Effect beyond bias	-0.104**	-0.036	-0.139***	-0.272	-0.278***
	(0.046)	(0.12)	(0.019)	(0.503)	(0.037)
Observations	669	669	665	669	669

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Table B4: Publication bias tests for subsets of data—cont.

Block 5: Ordinary least squares					
Panel A: Linear	OLS	FE	IV	Study	Precision
Publication bias	0.163**	0.241***	0.576*	-0.855***	0.383***
(*standard error*)	(0.0693)	(0.0579)	(0.313)	(0.238)	(0.133)
	[-0.680, 1.052]		[-0.00451, 1.211]	[-1.291, 0.172]	[0.0402, 1.071]
			{-0.0127, 1.288}		
Effect beyond bias	0.228	0.153**	-0.157	0.416*	0.0135
(*constant*)	(0.153)	(0.0565)	(0.236)	(0.249)	(0.0853)
	[-0.106, 0.607]		[-0.851, 0.295]	[-0.0916, 0.933]	[-0.294, 0.335]
First-stage robust F-stat			21.8		
Panel B: Nonlinear	WAAP	Stem	Kink	p-uniform*	Selection
Publication bias			0.615***		P = 0.842
			(0.151)		(0.239)
Effect beyond bias	-0.007	-0.003	-0.048***	0.342	0.007
	(0.009)	(0.039)	(0.018)	(0.238)	(0.061)
Observations	433	433	429	433	433

Notes: Panel A reports the results of a linear regression: $\mathrm{e}_{ij} = \mathrm{e}_0 + \beta \cdot SE(\mathrm{e}_{ij}) + \epsilon_{ij}$, where e_{ij} denotes the i-th class size effect estimated in the j-th study, and $SE(\mathrm{e}_{ij})$ denotes the standard error. The class size effects are normalized to represent the change in hundredths of a standard deviation in test scores corresponding to an increase in class size by one student. FE: study-level fixed effects. IV: reported standard errors are instrumented by the inverse of the square root of sample size. Study: estimates are weighted by the inverse of the number of estimates reported per study. Precision: estimates are weighted by their inverse variance. In Panel B, WAAP denotes the weighted average of adequately powered estimates (Ioannidis *et al.*, 2017), Stem denotes the stem-based technique (Furukawa, 2021), Kink denotes the endogenous kink model (Bom & Rachinger, 2019), p-uniform* denotes the technique due to van Aert & van Assen (2023), and Selection denotes the technique due to Andrews & Kasy (2019). In the selection model, P denotes the probability that estimates insignificant at the 5% level are published relative to the probability that significant estimates are published. Standard errors, clustered at the study level, are reported in parentheses. In square brackets we report 95% confidence intervals from wild bootstrap (Roodman *et al.*, 2018). For IV, in curly brackets we report the 95% confidence interval based on Anderson & Rubin (1949).* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B5: Publication bias tests with different definitions of the class size effect

Block 1: Effect of a one-standard-deviation increase in class size					
Panel A: Linear	OLS	FE	IV	Study	Precision
Publication bias	0.0757	0.0730	0.149	-0.0291	0.147
(*standard error*)	(0.0979)	(0.0799)	(0.150)	(0.0670)	(0.137)
	[-0.493, 0.231]		[-0.325, 0.565]	[-0.390, 0.052]	[-0.435, 0.408]
			{-0.163, 0.432}		
Effect beyond bias	-0.0706***	-0.0698***	-0.0943***	-0.138***	-0.0603***
(*constant*)	(0.0258)	(0.0237)	(0.0205)	(0.0470)	(0.0189)
	[-0.120, -0.0177]		[-0.169, -0.016]	[-0.235, -0.044]	[-0.133, -0.011]
First-stage robust F-stat			34.0		
Panel B: Nonlinear	WAAP	Stem	Kink	p-uniform*	Selection
Publication bias			-0.397***		P = 0.825
			(0.074)		(0.121)
Effect beyond bias	-0.004***	-0.007	-0.014***	-0.089**	-0.050***
	(0.001)	(0.040)	(0.002)	(0.041)	(0.007)
Observations	2,231	2,231	2,223	2,231	2,231

Block 2: Effects recomputed to partial correlation coefficients					
Panel A: Linear	OLS	FE	IV	Study	Precision
Publication bias	-0.890***	-0.579***	0.159	-0.722***	-0.267
(*standard error*)	(0.0480)	(0.0784)	(0.528)	(0.216)	(0.304)
	[-4.497, 0.281]		[-0.072, 0.389]	[-1.295, 0.131]	[-0.953, 0.395]
			{-0.417, 2.930}		
Effect beyond bias	0.00511	-0.00783**	-0.0387	-0.00516	-0.00372**
(*constant*)	(0.00572)	(0.00327)	(0.0460)	(0.00540)	(0.00145)
	[-0.007, 0.016]		[-0.052, -0.025]	[-0.016, 0.005]	[-0.0128, -0.001]
First-stage robust F-stat			8.9		
Panel B: Nonlinear	WAAP	Stem	Kink	p-uniform*	Selection
Publication bias			-0.267***		P = 0.660
			(0.091)		(0.065)
Effect beyond bias	-0.003**	-0.010	-0.004***	-0.007	-0.004***
	(0.001)	(0.006)	(0.0004)	(0.004)	(0.001)
Observations	2,819	2,819	2,817	2,819	2,819

Notes: This table use an alternative definition of effect size (Block 1) and recomputes effects to a simplified standardized metric that allows us to include studies not included in the previous analysis (Block 2). Note that partial correlations and their standard errors are correlated by construction. When we instrument standard errors, the relationship disappears. Panel A reports the results of a linear regression: $\mathrm{e}_{ij} = \mathrm{e}_0 + \beta \cdot SE(\mathrm{e}_{ij}) + \epsilon_{ij}$, where e_{ij} denotes the i-th class size effect estimated in the j-th study, and $SE(\mathrm{e}_{ij})$ denotes the standard error. In Block 1, class size effects are normalized to represent the change in hundredths of a standard deviation in test scores corresponding to an increase in class size by one standard deviation. In Block 2, class size effects are normalized to represent partial correlation coefficients. FE: study-level fixed effects. IV: reported standard errors are instrumented by the inverse of the square root of sample size. Study: estimates are weighted by the inverse of the number of estimates reported per study. Precision: estimates are weighted by their inverse variance. In Panel B, WAAP denotes the weighted average of adequately powered estimates (Ioannidis *et al.*, 2017), Stem denotes the stem-based technique (Furukawa, 2021), Kink denotes the endogenous kink model (Bom & Rachinger, 2019), p-uniform* denotes the technique due to van Aert & van Assen (2023), and Selection denotes the technique due to Andrews & Kasy (2019). In the selection model, P denotes the probability that estimates insignificant at the 5% level are published relative to the probability that significant estimates are published. Standard errors, clustered at the study level, are reported in parentheses. In square brackets we report 95% confidence intervals from wild bootstrap (Roodman *et al.*, 2018). For IV, in curly brackets we report the 95% confidence interval based on Anderson & Rubin (1949).* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B6: Andrews & Kasy (2019) model with multiple significance thresholds

Panel A: All estimates	Effect	Publication probabilities		
All journals (N = 2,434)	μ	[0, 1.645]	(1.645 , 1.96]	(1.96 , 2.576]
estimate	-0.225	0.605	0.687	0.780
standard error	0.013	0.050	0.075	0.073
clustered SE	0.046	0.176	0.264	0.257
Panel B: Preferred estimates				
All journals (N = 444)	μ	[0, 1.645]	(1.645 , 1.96]	(1.96 , 2.576]
estimate	-0.247	0.812	0.808	0.944
standard error	0.066	0.256	0.262	0.262
clustered SE	0.087	0.336	0.344	0.344

Notes: N = number of estimates. Clustered SE = standard error clustered at the study level. Effect = publication-bias corrected mean effect of class size on student achievement, normalized to represent the change in hundredths of a standard deviation in test scores corresponding to an increase in class size by one student. That is, an estimate of -1 means that a class size reduction by 10 students is associated with an improvement in test scores by 0.1 standard deviations. Brackets show intervals for reported t-statistics in absolute value. Publication probability is set to 1 for estimates significant at the 1% level.

Table B7: Diagnostics of the benchmark BMA estimation (UIP and dilution priors)

Mean no. regressors	*Draws*	*Burn-ins*	*Time*	*No. models visited*
12.6312	$3 \cdot 10^5$	$1 \cdot 10^5$	1.43 mins	48,370
Model space	*Visited*	*Top models*	*Corr PMP*	*No. obs.*
$2.7 \cdot 10^{11}$	0.0018%	100%	0.9970	2,434
Model prior	*g-prior*	*Shrinkage-stats*		
Random/19	UIP	Av = 0.9996		

Notes: We employ the combination of unit information prior recommended by (Eicher *et al.*, 2011) and dilution prior suggested by George (2010), which accounts for collinearity.

Figure B6: Benchmark BMA model size and convergence (UIP and dilution priors)

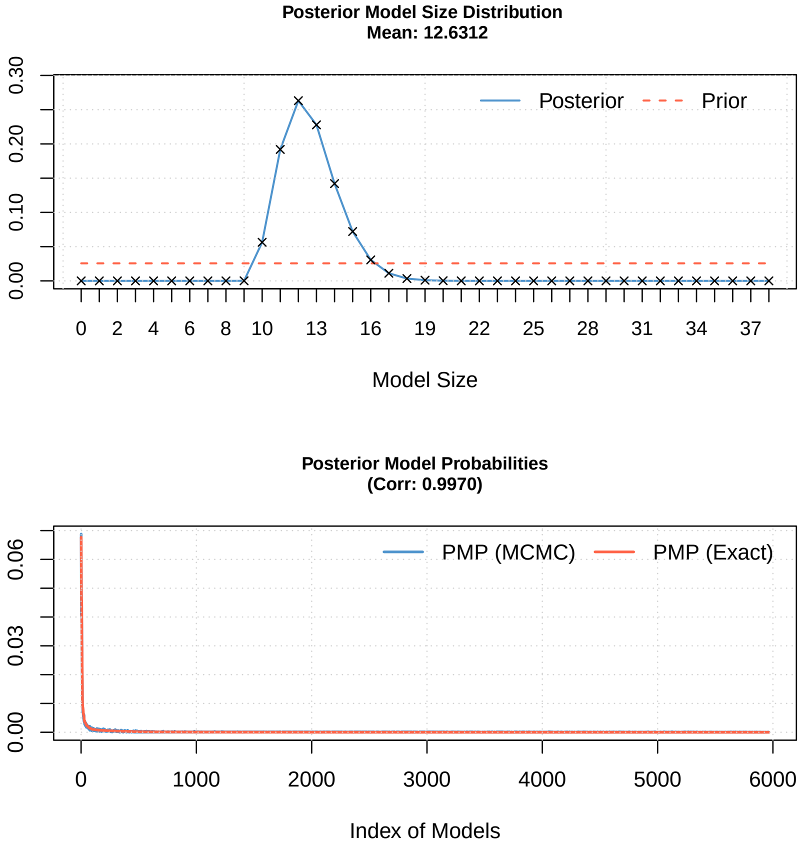
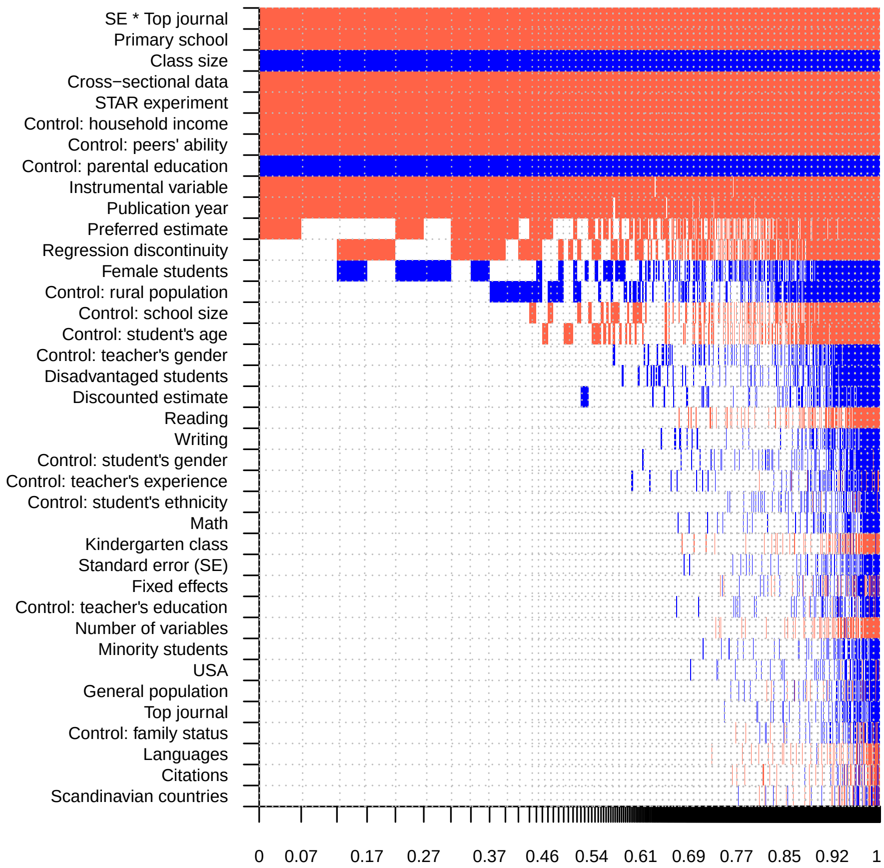


Table B8: Why estimates of the class size effect vary (robustness checks)

Response variable: class size effect	Bayesian model averaging BRIC and random priors			Frequentist model averaging		
	P. mean	P. SD	PIP	Coef.	SE	p-value
Constant	1.36	NA	1.00	0.90	1.18	0.45
Standard error (SE)	0.00	0.00	0.01	0.00	0.03	1.00
SE * Top journal	-1.55	0.18	1.00	-1.42	0.27	0.00
Subjects tested						
Math	0.00	0.02	0.02	0.00	0.07	1.00
Reading	-0.01	0.04	0.03	0.00	0.22	1.00
Writing	0.01	0.09	0.03	0.00	0.42	1.00
Languages	0.00	0.01	0.01	0.00	0.10	1.00
Class characteristics						
Kindergarten class	0.00	0.04	0.01	0.00	0.36	1.00
Primary school	-0.64	0.10	1.00	-0.62	0.13	0.00
Class size	0.59	0.10	1.00	0.59	0.14	0.00
Female students	0.34	0.46	0.39	0.45	0.61	0.46
Minority students	0.00	0.04	0.01	0.00	0.71	1.00
Disadvantaged students	0.02	0.08	0.06	0.08	0.65	0.90
General population students	0.00	0.02	0.01	0.00	0.49	1.00
Data characteristics						
Cross-sectional data	-1.04	0.15	1.00	-0.89	0.47	0.06
USA	0.00	0.02	0.01	0.00	0.13	1.00
Scandinavian countries	0.00	0.01	0.01	0.00	0.22	1.00
Estimation technique						
STAR experiment	-1.51	0.29	1.00	-1.42	0.57	0.01
Regression discontinuity	-0.19	0.22	0.48	-0.24	0.28	0.38
Instrumental variable	-0.43	0.12	0.99	-0.40	0.18	0.03
Fixed effects	0.00	0.02	0.01	0.00	0.05	1.00
Number of variables	0.00	0.01	0.01	0.00	0.17	1.00
Control: student's gender	0.01	0.05	0.03	0.00	0.48	1.00
Control: student's age	-0.05	0.12	0.17	-0.08	0.54	0.88
Control: student's ethnicity	0.00	0.02	0.02	0.00	0.23	1.00
Control: household income	-0.75	0.12	1.00	-0.69	0.17	0.00
Control: parental education	0.67	0.13	1.00	0.57	0.28	0.04
Control: family status	0.00	0.01	0.01	0.00	0.31	1.00
Control: peers' ability	-0.74	0.12	1.00	-0.67	0.19	0.00
Control: teacher's experience	0.00	0.03	0.02	0.00	0.14	1.00
Control: teacher's gender	0.02	0.09	0.07	0.07	0.59	0.90
Control: teacher's education	0.00	0.02	0.01	0.00	0.13	1.00
Control: school size	-0.06	0.14	0.19	-0.19	0.26	0.48
Control: rural population	0.11	0.19	0.28	0.19	0.29	0.52
Publication characteristics						
Preferred estimate	-0.17	0.18	0.52	-0.21	0.17	0.22
Discounted estimate	0.01	0.05	0.05	0.00	0.04	1.00
Top journal	0.00	0.03	0.01	0.00	0.58	1.00
Citations	0.00	0.01	0.01	0.00	0.07	1.00
Publication year	-0.55	0.15	0.97	-0.42	0.30	0.16
Observations	2,434			2,434		

Notes: The response variable is the estimate of the effect of size class on student achievement. The class size effects are normalized to represent the change in hundredths of a standard deviation in test scores corresponding to an increase in class size by one student. SE = standard error, P. mean = posterior mean, P. SD = posterior standard deviation, PIP = posterior inclusion probability. In the first specification from the left we employ Bayesian model averaging (BMA) using BRIC g-prior suggested by Fernandez *et al.* (2001) and the beta-binomial model prior according to Ley & Steel (2009). The specification on the right employs frequentist model averaging by applying Mallows weights Hansen (2007) using orthogonalization of the covariate space suggested by Amini & Parmeter (2012) to reduce the number of estimated models. The posterior mean in Bayesian model averaging (or alternatively the estimated coefficient in frequentist model averaging) denotes the marginal effect of a study characteristic on the effect size reported in the literature. For detailed description of all the variables see Table 6.

Figure B7: Model inclusion in BMA (BRIC and random priors)



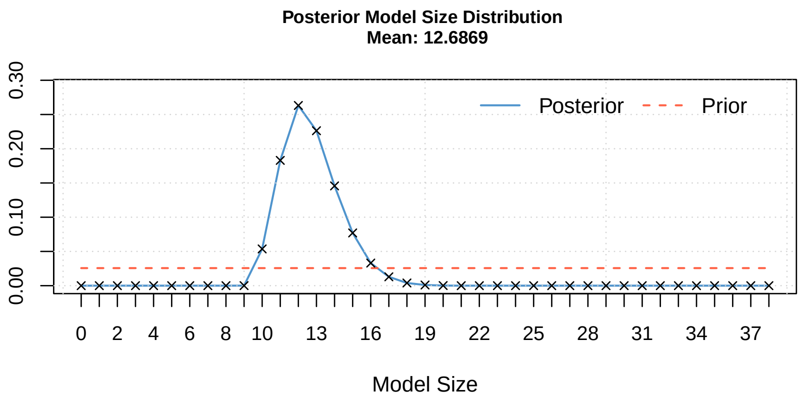
Notes: On the vertical axis the explanatory variables are ranked according to their posterior inclusion probabilities from the highest at the top to the lowest at the bottom. The horizontal axis shows the values of cumulative posterior model probability. Blue color (darker in grayscale) = the estimated parameter of a corresponding explanatory variable is positive. Red color (lighter in grayscale) = the estimated parameter of a corresponding explanatory variable is negative. No color = the corresponding explanatory variable is not included in the model. Numerical results are reported in Table B8. All variables are described in Table 6.

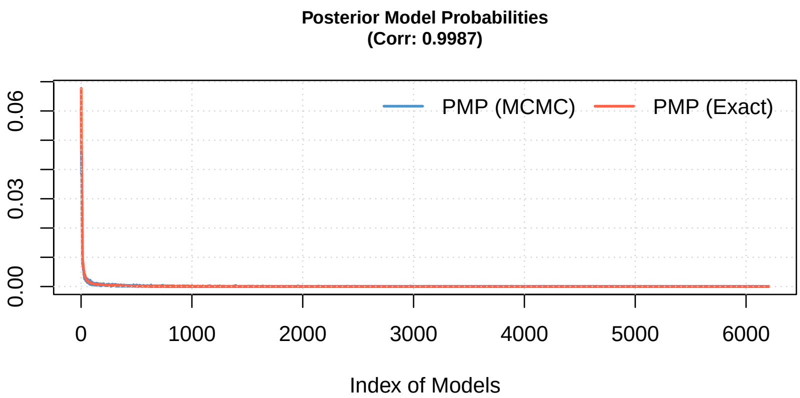
Table B9: Diagnostics of the BMA estimation (BRIC and random priors)

Mean no. regressors	*Draws*	*Burn-ins*	*Time*	*No. models visited*
12.6869	$3 \cdot 10^5$	$1 \cdot 10^5$	1.55mins	48,632
Model space	*Visited*	*Top models*	*Corr PMP*	*No. obs.*
$2.7 \cdot 10^{11}$	0.0018%	100%	0.9987	2,434
Model prior	*g-prior*	*Shrinkage-stats*		
Random/19	BRIC	Av = 0.9996		

Notes: The specification uses a BRIC *g*-prior suggested by Fernandez *et al.* (2001) and the beta-binomial model prior according to Ley & Steel (2009).

Figure B8: BMA model size and convergence (BRIC and random priors)





References

van Aert, R. C. & M. van Assen (2023): "Correcting for publication bias in a meta-analysis with the p-uniform* method." *Working paper*, Tilburg University & Utrecht University.

Amini, S. M. & C. F. Parmeter (2012): "Comparison of model averaging techniques: Assessing growth determinants." *Journal of Applied Econometrics* **27(5)**: pp. 870–876.

Anderson, T. W. & H. Rubin (1949): "Estimation of the Parameters of Single Equation in a Complete System of Stochastic Equations." *Annals of Mathematical Statistics* **20**: pp. 46–63.

Andrews, I. & M. Kasy (2019): "Identification of and correction for publication bias." *American Economic Review* **109(8)**: pp. 2766–2794.

Argaw, B. & P. Puhani (2018): "Does class size matter for school tracking outcomes after elementary school? Quasi-experimental evidence using administrative panel data from Germany." *Economics of Education Review* **65(C)**: pp. 48–57.

Bom, P. R. & H. Rachinger (2019): "A kinked meta-regression model for publication bias correction." *Research Synthesis Methods* **10(4)**: pp. 497–514.

Bressoux, P., L. Lima, & C. Monseur (2019): "Reducing the number of pupils in French first-grade classes: Is there evidence of contemporaneous and carryover effects?" *International Journal of Educational Research* **96(C)**: pp. 136–145.

Eicher, T. S., C. Papageorgiou, & A. E. Raftery (2011): "Default priors and predictive performance in Bayesian model averaging, with application to growth determinants." *Journal of Applied Econometrics* **26(1)**: pp. 30–55.

Elpais (2020): "Madrid will delay start of school term for some students and bring down class sizes." *Online article as of Aug 25, 2020*, Ediciones El Pais.

Evain, F. (2022): "Class size in primary education: The decline continues due to the impact of reducing the last grade of pre-primary classes." *Note d'information 22/08*, Ministere de l'Education nationale et de la Jeunesse, DEPP, France.

Fadulu, L. (2022): "Class Sizes Set to Shrink in New York City Schools, but at What Cost?" *NYT article as of Jun 3, 2022*, The New York Times.

Fernandez, C., E. Ley, & M. F. Steel (2001): "Benchmark priors for Bayesian model averaging." *Journal of Econometrics* **100(2)**: pp. 381–427.

FNAE (2019): "Average group sizes in basic education in Finland below the OECD average." *Online article as of Aug 8, 2019*, Finnish National Agency for Education, Finland.

Furukawa, C. (2021): "Publication bias under aggregation frictions: From communication model to new correction method." *MIT working paper*, Massachusetts Institute of Technology, Cambridge, MA.

George, E. I. (2010): "Dilution priors: Compensating for model space redundancy." In "IMS Collections Borrowing Strength: Theory Powering Applications – A Festschrift for Lawrence D. Brown," volume 6, p. 158–165. Institute of Mathematical Statistics.

Han, J. & K. Ryu (2017): "Effects of class size reduction in upper grades: Evidence from Seoul, Korea." *Economics of Education Review* **60(C)**: pp. 68–85.

Hansen, B. E. (2007): "Least squares model averaging." *Econometrica* **75(4)**: pp. 1175–1189.

Ioannidis, J. P., T. Stanley, H. Doucouliagos *et al.* (2017): "The power of bias in economics research." *The Economic Journal* **127(605)**: pp. 236–265.

JP (2015): "Ministers approve Bennett reform to reduce class sizes." *Online article as of Nov 16, 2015*, The Jerusalem Post.

Koreaherald (2016): "Korea to cut class size of high schools." *Online article as of Apr 25, 2016*, The Korea Herald.

Ley, E. & M. F. Steel (2009): "On the effect of prior assumptions in Bayesian model averaging with applications to growth regression." *Journal of Applied Econometrics* **24(4)**: pp. 651–674.

MECF (2012): "Quality Criteria in Basic Education (In Finnish: Perusopetuksen laatukriteerit. Perusopetuksen, perusopetuksen aamu- ja iltapaivatoiminnan seka koulun kerhotoiminnan laatukriteerit)." *Publications of the Ministry of Education and Culture 2012/29*, Ministry of Education and Culture, Finland.

MERN (2019): "Compulsory education facts and learning outcomes: The Norwegian Education Mirror, 2019." *Education Mirror 2019 online publicaton*, The Norwegian Directorate for Education and Training, Ministry of Education and Research, Norway.

NASUWT (2023): "Class Sizes." *SNCT Handbook as of May 2023*, NASUWT, National Association of Schoolmasters Union of Women Teachers, United Kingdom and SNCT, Scottish Negotiating Committee for Teachers, Scotland UK.

NIE (2020): "Not more than 30 students in a class, says New Education Policy." *Online article as of July 30, 2020*, The New Indian Express.

NSWG (2023): "History of New South Wales government schools." *Online article as of Mar 6, 2023*, New South Wales Government, Australia.

NSWTF (2019): "How many students should be in my class?" *Online source from Nov 10, Section: News*, New South Wales Teachers Federation, Australia.

OECD (2020): "Educational Policy Outlook: Portugal." *Educational Policy Outlook 7/2020*, OECD.

OME (2019): "Class size engagement guide." *Online memo as of Jan 2019*, Ontario Ministry of Education, Canada.

QME (2023): "Path 5 - Reduce the number of students per class in elementary school." *Online article as of May 11, 2023*, Quebec Ministry of Education and Ministry of Higher Education, Canada.

Roodman, D., J. G. MacKinnon, M. O. Nielsen, & M. D. Webb (2018): "Fast and wild: Bootstrap inference in Stata using boottest." *Queen's Economics Department Working Paper 1406*, Department of Economics, Queen's University, Canada: Kingston.

Rubio, S. (2022): "Class Size Reduction Bill." *Senator's announcement as of Apr 20, 2022*, Senate District 22, US Senate, United States.

Tinetti, H. J. (2023): "Smaller class sizes to improve teaching and learning outcomes." *The official government website as of April 17, 2023*, Ministry of Education, New Zealand Government.

VARC (2023): "An Integrated Qualitative and Quantitative Evaluation of the SAGE Project." *SAGE project web*, Value-Added Research Center at Wisconsin Center for Education Research at the School of Education, University of Wisconsin-Madison, United States.

WDPI (2016): "Achievement Gap Reduction (AGR) Program." *Strategy Resource Guide*, Wisconsin Department of Public Instruction, United States.

Whitehurst, G. J. & M. M. Chingos (2011): "Class Size: What Research Says and What it Means for State Policy." *Report 5/2011*, Brown Center on Education Policy at Brookings, Washington, DC.